

Tuple

Tuple Creation

```
In [1]: tup1 = ()    # Empty tuple
```

```
In [2]: tup2 = (10, 30, 60)    # Tuple of integer numbers
```

```
In [3]: tup3 = (10.77, 30.66, 60.89)    # Tuple of float numbers
```

```
In [4]: tup4 = ('one', 'two', 'three')    # Tuple of strings
```

```
In [5]: tup5 = ('Asif', 25, (50,100), (150, 90))    # Nested tuples - Tuple inside tuple
```

```
In [6]: tup6 = (100, 'Asif', 17.765)    # Tuple of mixed data types
```

```
In [7]: tup7 = ('Asif', 25, [50, 100], [150, 90], {'John', 'David'}, (99,22,33))
```

```
In [8]: len(tup7)    # Length of List
```

```
Out[8]: 6
```

Tuple Indexing

```
In [9]: tup2[0]    # Retrieve first element of the tuple
```

```
Out[9]: 10
```

```
In [10]: tup4[0]    # Retrieve first element of the tuple
```

```
Out[10]: 'one'
```

```
In [11]: tup4[0][0]    # Nested indexing - Access the first character of the first tuple element
```

```
Out[11]: 'o'
```

```
In [12]: tup4[-1]    # Last item of the tuple
```

```
Out[12]: 'three'
```

```
In [13]: tup5[-1]    # Last item of the tuple
```

```
Out[13]: (150, 90)
```

Tuple Slicing

```
In [14]: mytuple = ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [15]: mytuple[0:3]    # Return all items from 0th to 3rd index location excluding the i
```

```
Out[15]: ('one', 'two', 'three')
```

```
In [16]: mytuple[2:5]    # Return all items from 2nd to 5th index location excluding the i
```

```
Out[16]: ('three', 'four', 'five')
```

```
In [17]: mytuple[:3]     # Return first three items
```

```
Out[17]: ('one', 'two', 'three')
```

```
In [18]: mytuple[:2]     # Return first two items
```

```
Out[18]: ('one', 'two')
```

```
In [19]: mytuple[-3:]    # Return last three items
```

```
Out[19]: ('six', 'seven', 'eight')
```

```
In [20]: mytuple[-2:]    # Return last two items
```

```
Out[20]: ('seven', 'eight')
```

```
In [21]: mytuple[-1:]    # Return last item of the tuple
```

```
Out[21]: ('eight',)
```

```
In [22]: mytuple[:]      # Return whole tuple
```

```
Out[22]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

Remove & Change items

```
In [23]: mytuple
```

```
Out[23]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [24]: del mytuple[0]    # Tuples are immutable which means we can't DELETE tuple items
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[24], line 1
----> 1 del mytuple[0]    # Tuples are immutable which means we can't DELETE tuple
items

TypeError: 'tuple' object doesn't support item deletion
```

```
In [25]: mytuple[0] = 1    # Tuples are immutable which means we can't CHANGE tuple items
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[25], line 1  
----> 1 mytuple[0] = 1    # Tuples are immutable which means we can't CHANGE tuple  
items  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [26]: del mytuple    # Deleting entire tuple object is possible
```

Loop through a tuple

```
In [27]: mytuple
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[27], line 1  
----> 1 mytuple  
  
NameError: name 'mytuple' is not defined
```

```
In [28]: mytuple = ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [29]: mytuple
```

```
Out[29]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [30]: for i in mytuple:  
         print(i)
```

```
one  
two  
three  
four  
five  
six  
seven  
eight
```

```
In [31]: for i in enumerate(mytuple):  
         print(i)
```

```
(0, 'one')  
(1, 'two')  
(2, 'three')  
(3, 'four')  
(4, 'five')  
(5, 'six')  
(6, 'seven')  
(7, 'eight')
```

Tuple Membership

```
In [32]: mytuple
```

```
Out[32]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [33]: 'one' in mytuple    # Check if 'one' exist in the list
```

```
Out[33]: True
```

```
In [34]: 'ten' in mytuple    # Check if 'ten' exist in the list
```

```
Out[34]: False
```

```
In [35]: if 'three' in mytuple:    # Check if 'three' exist in the list
          print('Three is present in the table')
        else:
          print('Three is not present in the table')
```

Three is present in the table

```
In [36]: if 'eleven' in mytuple:    # Check if 'eleven' exist in the list
          print('Eleven is present in the table')
        else:
          print('Eleven is not present in the table')
```

Eleven is not present in the table

Index Position

```
In [37]: mytuple
```

```
Out[37]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [38]: mytuple.index('one')    # Index of first element equal to 'one'
```

```
Out[38]: 0
```

```
In [39]: mytuple.index('five')    # Index of first element equal to 'five'
```

```
Out[39]: 4
```

```
In [40]: mytuple1 = ('one', 'two', 'three', 'four', 'one', 'one', 'two', 'three')
```

```
In [41]: mytuple1
```

```
Out[41]: ('one', 'two', 'three', 'four', 'one', 'one', 'two', 'three')
```

```
In [42]: mytuple1.index('one')    # Index of first element equal to 'one'
```

```
Out[42]: 0
```

Sorting

```
In [43]: mytuple2 = (43, 67, 99, 12, 6, 90, 67)
```

```
In [44]: sorted(mytuple2)    # Returns a new sorted list and doesn't change original tuple
```

```
Out[44]: [6, 12, 43, 67, 67, 90, 99]
```

```
In [45]: sorted(mytuple2, reverse = True)    # Sorting in descending order
```

Out[45]: [99, 90, 67, 67, 43, 12, 6]

In []: